ORIGINAL ARTICLE

Systems engineering; model-based systems engineering; nanosatellites; electrical power systems

Model-Based Systems Engineering for Nanosatellite Electrical Power Systems: Generality, Value Claims, and Pre-Physical Verification

Author Information To Be Supplied[1]

1 | METHODOLOGY

1.1 | Protocol and research questions

This review used a predefined protocol to delimit the population, intervention, outcomes, eligibility criteria, and synthesis approach. The protocol defines a nanosatellite-class population, an MBSE intervention, and outcomes concerned with general modeling, EPS-relevant value, and verification before physical development. The review method is intentionally narrower than broad space-MBSE syntheses: Buruso et al. Buruso et al. (2026) are used as a methodological and positioning comparator, especially for transparent review procedure and concise enabler classification, but their review is not treated as evidence about the studies included here.

The protocol research questions were:

RQ1. How can nanosatellite missions and EPS be modeled in a general manner with minimal knowledge about their behavior?

RQ2. What are the claimed outcomes related to reliability, reproducibility, and integrability derived from EPS modeling in nanosatellite missions?

RQ3. How have verification procedures been integrated into EPS models prior to physical development?

The population was interpreted as CubeSat and nanosatellite-class missions rather than larger spacecraft or smaller educational platforms such as CanSats. For RQ1, the focal intervention was system-level or EPS-level MBSE. For RQ2, outcomes had to be attributable to MBSE artifacts, MBSE activities, or model-enabled process changes rather than to the selected EPS technology alone. For RQ3, the review looked for requirements, verification, validation, IVV, test, simulation, or analysis artifacts integrated in or around the model before hardware development.

Abbreviations: EPS, electrical power system; MBSE, model-based systems engineering; PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses; RQ, research question; SIL, software-in-the-loop; HIL, hardware-in-the-loop; V&V, verification and validation

1.2 | Information sources and search strategy

Records were retrieved from Scopus and IEEE Xplore. Scopus searches were run directly, while IEEE Xplore records were exported and screened using the same eligibility criteria. Targeted exact-title Google Scholar searches were used only to verify or recover known titles identified through citation chasing; Google Scholar was not used as an open-ended third database.

Search strings combined nanosatellite/CubeSat population terms with MBSE terms such as "model-based systems engineering", MBSE, SysML, and selected alternative terminology including Capella, Arcadia, and cyber-physical systems where relevant. The search strategy followed four concept clusters derived from the protocol: (i) general nanosatellite or CubeSat MBSE reference models, templates, behavior models, and mission/system architectures; (ii) EPS, power-subsystem, power-budget, and electrical-architecture modeling; (iii) verification, validation, IVV, test, requirements-in-the-loop, and traceability; and (iv) outcome and integration terms such as reliability, reproducibility, integrability, interoperability, and lifecycle continuity. The final record set combined database search results, eligible IEEE Xplore records, and targeted citation-chasing recoveries.

Search and screening limits followed the protocol: publication years 2013–2026, English language, conference or journal papers, and either open access or availability through institutional library access. Search details were recorded for query identifiers, query strings, result totals, recovery searches, and stopping rationale. The stopping decision was based on saturation of the Scopus–IEEE evidence base and on the observation that later EPS/V&V recall passes mainly confirmed sparsity rather than uncovering a large hidden literature.

1.3 | Selection process and eligibility criteria

Records from the retained searches were merged and deduplicated before screening. The final PRISMA-style flow was: 199 records identified, 118 duplicates removed, 81 unique records screened, 37 records excluded during screening, 44 reports sought for retrieval, 4 reports not retrieved or inaccessible, and 40 reports extracted.

Screening applied the protocol criteria as follows:

- include English conference or journal papers from 2013–2026;
- include CubeSat, nanosatellite, or very close nanosatellite-class system-design studies, and exclude larger generic spacecraft, payload-only, ground-segment-only, software-only, hardware-calibration, education/tutorial, proceedings-volume, and non-space records unless a specific protocol-based rationale supported retention;
- include studies using MBSE, SysML, model-based architecture, model-based requirements, model-based simulation/analysis, or closely related digital-engineering approaches at system level, mission level, EPS level, or an EPS-adjacent power/electrical-architecture level;
- for RQ1, prioritize system-level or EPS-level models, reference models, templates, reusable workflows, behavior models, and architecture models over isolated subsystem models;
- for RQ2, retain claimed outcomes only when the outcome could be linked to MBSE artifacts or model-enabled processes, not merely to EPS component choices;
- for RQ3, require explicit verification, validation, IVV, test, requirements traceability, simulation, parametric checking, or pre-physical-development evaluation evidence in or around the model.

Screening distinguished core and supporting studies. Core papers directly shaped the synthesis because they were central to CubeSat/nanosatellite MBSE, EPS/power modeling, traceability, reusable reference models, or V&V.

Supporting papers were retained when they provided contextual evidence on toolchains, lifecycle continuity, requirements-in-the-loop, risk, project-management, or related nanosatellite MBSE practices without being as central to the protocol RQs.

1.4 | Data extraction and quality appraisal

Extraction was performed on the 40 accessible reports, comprising 21 core and 19 supporting studies. The four reports that remained inaccessible at retrieval were classified as unavailable and excluded from the evidence synthesis.

A standardized extraction form was used for every accessible study and was refactored into two linked layers. First, a controlled codebook was derived from the protocol and used for synthesis-critical fields with explicit allowed values, including RQ relevance, lifecycle stage, EPS relevance, model maturity, modeling scope, MBSE approach classification, toolchain mechanisms, EPS/V&V artifacts, IVV integration level, outcome categories, reliability subtype, evidence types, gap mapping, quality ratings, limitation flags, and extraction confidence. Second, open extraction preserved researcher observations, unexpected mechanisms, tools, claims, caveats, and emergent themes that did not fit the controlled vocabulary. Extracted fields covered bibliographic metadata; research-question relevance; mission/platform context; lifecycle stage; EPS relevance; model maturity; MBSE objective, scope, engineering artifacts, abstraction level, and reuse mechanism; implementation pathway and tailoring; toolchain, interoperability, and lifecycle-continuity mechanisms; EPS and verification-specific evidence; claimed outcomes, mechanisms, evidence types, metrics, baselines, and confidence ratings; governance, ROI, deliverables, and project-decision linkage; gap mapping; limitations; supporting quotations; lineage notes; and extraction confidence.

Each per-study extraction therefore contains both a compact controlled-code summary for aggregation and open 'emergent codes' for qualitative synthesis. Controlled codes were assigned only when the surrounding meaning supported the classification; keyword presence alone was not treated as evidence. After the codebook was refined, each paper received a focused per-study re-coding pass for generality/reuse, IVV integration level, and reliability subtype so that these later fields were reasoned from the paper context rather than inferred from previous summaries. Values coded as "other" in controlled fields were retained as "other" and explained through emergent codes rather than silently expanding the codebook. Limitation flags were used only for aggregation, while author-stated and extractor-identified limitation rationales were retained as text.

Quality appraisal used the protocol checklist rather than an exclusion threshold. Each checklist item was scored as No = 0, Insufficiently = 0.5, or Yes = 1. The checklist assessed venue credibility; whether limitations, threats, assumptions, or scope boundaries were discussed; clarity of nanosatellite/CubeSat context and lifecycle stage; clarity of MBSE method, language, toolchain, model artifacts, and system boundary; whether claimed outcomes were explicitly linked to MBSE artifacts or activities; whether evidence was provided for claims; whether requirements, verification, validation, design decisions, or lifecycle gates were traced to model artifacts when relevant; and whether model availability, reusable assets, data exchange, or implementation details were reported. Quality scores were used to inform sensitivity, confidence, and discussion of evidence strength; they were not used to automatically remove studies after full-text extraction.

No independent inter-rater reliability assessment was performed. Consequently, the review does not report agreement statistics. Instead, extraction consistency was supported by the fixed protocol, a common extraction form, explicit rationales and direct quotes for coding decisions, extraction-confidence ratings, cross-checking extracted data against source reports and eligibility decisions, and the focused per-study re-coding pass for fields introduced by the codebook refinement.

1.5 | Synthesis strategy

The synthesis was structured by the three protocol RQs rather than by a general MBSE tool or method taxonomy. Descriptive statistics were used to summarize study characteristics, study role (core or supporting), EPS relevance, approach classifications, quality-score distribution, research-question relevance, value-claim evidence, and V&V linkage categories. Narrative synthesis then interpreted these counts in relation to the protocol questions.

A concise enabler taxonomy was retained only as a secondary explanatory layer. The Pure, Enhanced, Extended, and Integrated MBSE/digital-engineering categories, together with lifecycle activity anchors such as requirements management, model development, simulation/analysis, configuration management, and collaboration/communication, were used as sensitizing categories consistent with the broader space-MBSE review style of Buruso et al. Buruso et al. (2026). The main synthesis, however, focused on two issues specific to this review: the evidence behind claimed value and the linkage between MBSE artifacts and V&V before physical development.

For RQ1, studies were compared by modeling focus: whole-nanosatellite models that include EPS or power as a subsystem, generic whole-nanosatellite models without explicit EPS, and EPS-only or EPS-first models. Reuse mechanisms such as reference models, templates, profiles, libraries, open packages, repeatable workflows, high-level abstractions, mission-agnostic architectures, interface-focused models, low-dynamics or non-detailed-behavior models, and specific-mission models with reusable methods were examined as evidence for generality and for modeling with limited prior behavioral knowledge. For RQ2, each value claim was synthesized by controlled outcome category, reliability subtype, specificity, MBSE mechanism, evidence type, metric or value indicator, baseline or comparator, and evidence-confidence rating. Multi-label outcome-domain counts were derived from controlled outcome codes only, not from loose keyword searches over claim text; open observations were retained separately as emergent codes for qualitative interpretation. This prevented asserted benefits such as reliability, reproducibility, integrability, schedule reduction, or communication improvement from being treated as measured effects unless the extraction recorded supporting evidence. For RQ3, V&V linkage was synthesized from controlled V&V artifact, pre-physical verification, integration/test-readiness, and IVV integration-level codes, including requirements-to-model traceability, verification matrices, parametric and power-budget checks, executable simulation or scenario analysis, model-contained verification artifacts, test cases and procedures, automated tests, external tool/data-flow coupling, and evidence of physical, software-in-the-loop, hardware-in-the-loop, or integration-test readiness.

1.6 | Review validity

Several measures were used to strengthen review validity while preserving the limits of the available evidence base. Construct validity was supported by deriving eligibility, extraction fields, and synthesis categories from the protocol RQs and by keeping the broad space-MBSE review Buruso et al. (2026) as a positioning reference rather than as evidence about the included studies. Search validity was supported by using two major engineering bibliographic sources, exact query logging, deduplication, targeted recall searches for EPS/V&V sparsity, IEEE parity checks, and citation-chasing recovery for specific missing titles. The main search limitation is that the database scope was restricted to Scopus and IEEE Xplore, with Google Scholar used only for targeted exact-title verification, so grey literature and some publisher-specific records may be underrepresented.

Selection validity was supported by explicit inclusion/exclusion criteria and recorded screening decisions. Residual uncertainty remains for records with limited abstracts, borderline eligibility decisions, and inaccessible reports. Extraction validity was supported by full-text extraction for accessible studies, supporting quotations, quality appraisal, claim-evidence coding, and extraction-confidence ratings. The absence of an independent inter-rater assessment is a

limitation; no reliability statistic is claimed.

Synthesis validity was addressed by separating descriptive study patterns from evidence-backed findings. Repeated publications from the same CubeSat reference-model lineage were treated with lineage notes so that publication count would not be mistaken for independent replication. Claims about MBSE value were interpreted through evidence confidence and baseline availability, and EPS-specific conclusions were kept conservative because the evidence base contains many system-level or EPS-adjacent studies but few direct EPS-first studies. External validity is therefore limited to nanosatellite/CubeSat-class MBSE under the search scope and should not be generalized to all space MBSE practice or to all spacecraft EPS development without further evidence.

2 | RESULTS

2.1 | Search, screening, retrieval, and extraction flow

The final synthesis included the 40 accessible studies that underwent standardized full-text extraction. The review flow was: 199 records identified, 118 duplicates removed, 81 unique records screened, 37 records excluded during screening, 44 reports sought for retrieval, 4 reports not retrieved or inaccessible, and 40 reports extracted. Unretrieved reports were excluded from the evidence synthesis.

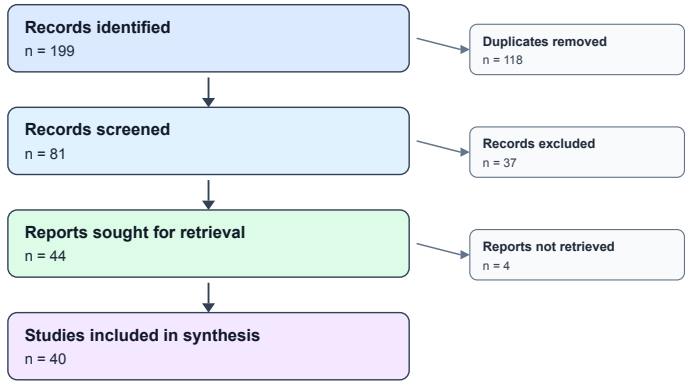


FIGURE 1 Flow of record identification, screening, retrieval, and inclusion for the review synthesis.

The extracted set comprised 21 core studies and 19 supporting studies published from 2013 to 2025. Counts below therefore refer to the 40 extracted studies unless identified as multi-label claim-domain or V&V-linkage counts.

2.2 | Study characteristics and quality

The quality checklist scores ranged from 5.0 to 7.5 on the eight-item scale, with a mean of 6.21. The score distribution was 5.0 ($n = 3$), 5.5 ($n = 6$), 6.0 ($n = 13$), 6.5 ($n = 9$), 7.0 ($n = 7$), and 7.5 ($n = 2$). Extraction confidence was coded as high for 29 studies and medium for 11 studies. RQ relevance was uneven: RQ1 was high for 24 studies and partial for 16; RQ2 was high for 3 and partial for 37; and RQ3 was high for 2, partial for 29, and not relevant for 9. The mostly

partial RQ2 pattern reflects that most papers stated MBSE value claims but did not directly measure EPS reliability, reproducibility, or integrability outcomes.

TABLE 1 Study, quality, and claim-evidence summary.

Metric	n
Extracted studies	40
Core studies	21
Supporting studies	19
Publication year range	2013–2025
Claim records	270
Unsupported claims	52
Weak claims	139
Moderate claims	77
Strong claims	2
High-confidence extractions	29
Medium-confidence extractions	11

Structured limitation flags further contextualize quality. The dominant limitations were incomplete lifecycle coverage ($n = 40$), weak claim–evidence linkage ($n = 40$), single-case or limited-scope evidence ($n = 40$), no independent validation ($n = 40$), prototype or future-work maturity ($n = 40$), no physical-test or flight evidence ($n = 39$), no baseline or comparator ($n = 38$), no quantitative evaluation ($n = 35$), unavailable model assets ($n = 34$), no EPS-specific evidence ($n = 33$), interoperability limitations ($n = 29$), and no pre-physical verification model ($n = 20$). These flags were treated as evidence-strength constraints rather than as exclusion criteria, and open limitation text was retained separately from the aggregation flags.

2.3 | Modeling focus and EPS relevance

Modeling focus and EPS relevance were synthesized as a single mutually exclusive classification. Thirty-four studies contained some EPS or power relevance: 19 reported concrete power-budget or electrical-architecture artifacts, 13 were EPS-adjacent, and 2 treated the EPS itself as the primary model scope. The remaining 6 studies did not model EPS explicitly: 4 were system-level nanosatellite or CubeSat MBSE studies without power-specific artifacts, and 2 contributed nanosatellite MBSE or V&V evidence outside the EPS domain.

The 19 power-budget/electrical-architecture studies should therefore be read as studies with concrete power-related evidence, not as 19 EPS-centered MBSE studies. Most of this evidence was embedded in system-level models or preliminary design studies. Examples include RAX/CubeSat energy and data simulations Kaslow et al. (2014); Spangelo et al. (2013); Anderson et al. (2014), SysML parametric mass and power-margin checks Waseem and Sadiq (2018), Capella/Arcadia preliminary EPS allocation for 3ColStar Buitrago-Leiva et al. (2024), CNES concurrent electrical-architecture simulations Boan (2025), and PLM/Dymola requirements-in-the-loop power simulations Rehman and Fortin (2024). The two EPS-specific studies were the open-source Papyrus/SysML power-subsystem

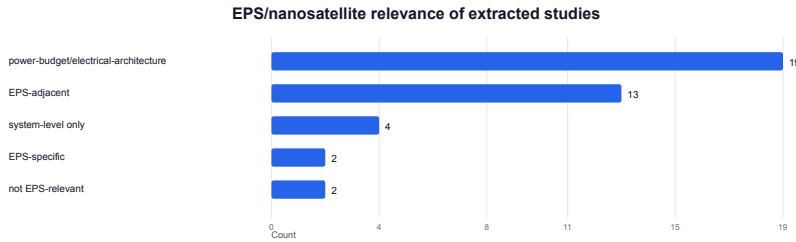


FIGURE 2 Mutually exclusive EPS relevance categories in the included studies.

TABLE 2 Modeling focus and EPS relevance categories.

EPS relevance category	n
Power-budget/electrical-architecture	19
EPS-adjacent	13
System-level only	4
EPS-specific	2
Not EPS-relevant	2

model with Java-based parameter and requirement-traceability extraction Strobbe et al. (2023) and the requirements-in-the-loop PLM/Dymola CubeSat case in which EPS behavior and state-of-charge simulation were central Rehman and Fortin (2024). Not-EPS-relevant studies still contributed general nanosatellite MBSE or V&V evidence, such as the EIRSAT-1 communications-subsystem verification demonstrator Akarapu et al. (2023).

This synthesis is intentionally stricter than simply counting any paper that mentioned power or included a generic EPS block. EPS appeared frequently as a system concern, but direct EPS-first MBSE evidence remained sparse even after the refactored extraction separated EPS-specific, EPS-adjacent, and power-budget evidence.

Generality and reuse were also multi-label. Method guidance appeared in 39 studies and reusable workflows in 34, while the refined generality codes identified high-level abstraction in 33 studies, mission-agnostic architecture in 22, reference models in 21, specific-mission models with reusable methods in 14, interface-focused models in 12, libraries in 11, and low-dynamics or non-detailed-behavior models in 10. These counts show that most papers supported generality through guidance, templates, or abstract architectures rather than through a released reusable EPS model. Only 6 studies reported open packages, and only 3 were coded as pattern-based reuse.

2.4 | MBSE approaches, languages, and toolchains

Approach classification was dominated by extended and enhanced MBSE rather than fully integrated digital-engineering environments. Seventeen studies were coded as Extended MBSE, 11 as Enhanced MBSE, 6 as Integrated MBSE/digital engineering, and 6 as Pure MBSE. This distribution situates the evidence base but does not by itself establish value or V&V effectiveness.

Tool and language classification showed a SysML-centered set of studies: SysML appeared in 28 studies, custom

TABLE 3 MBSE approach-classification counts.

Approach classification	n
Extended MBSE	17
Enhanced MBSE	11
Integrated MBSE/digital engineering	6
Pure MBSE	6

tools in 17, Cameo/MagicDraw/CATIA Magic in 14, MATLAB/Simulink and STK in 10 each, open-source packages in 9, ModelCenter and PLM/ALM tools in 6 each, Python in 5, Capella/Arcadia in 4, requirements tools in 4, and OpenMBEE in 3. The high count for "other" tools in the controlled summary reflects named toolchains that were preserved as emergent codes rather than forced into loose categories. Results-relevant toolchains included Papyrus/Eclipse for the EPS-specific open-source case Strobbe et al. (2023), Gaphor/UML stereotypes for open-source preliminary-design formalization Luccisano et al. (2024), docs-as-code/M30ML for SeaLion documentation and traceability Chiu et al. (2023), Dymola/Modelica in 3DExperience Rehman and Fortin (2024), and IDM-CIC/Simu-CIC/Opalis for electrical-architecture concurrent engineering Boan (2025).

Toolchain use was usually reported as a means to connect requirements, architecture, analysis, and verification artifacts. The strongest toolchain evidence came from executable or data-flow demonstrations, such as MagicDraw/Cameo–ModelCenter–STK–MATLAB integrations for RAX-style CubeSat analysis Kaslow et al. (2014); Spangelo et al. (2013), Capella/ForPlan or Capella/open-source simulation coupling de Almeida et al. (2021); Campioli et al. (2025), and SysML/PLM/Dymola requirements-in-the-loop simulation Rehman and Fortin (2024). Pure reference-model studies reported reusable structures and V&V packages, but generally provided less executed analysis evidence Kaslow and Madni (2017); Kaslow et al. (2019).

2.5 | Claimed outcomes and value-evidence strength

We extracted 270 claim records. Evidence confidence for those claims was classified as Unsupported ($n = 52$), Weak ($n = 139$), Moderate ($n = 77$), and Strong ($n = 2$). The two strong claims both came from a single concurrent-model-based CubeSat mission-design study that reported a quantitative comparison with a traditional feasibility-study process Bella et al. (2024). Outcome domains are multi-label and are now derived from controlled outcome codes only; the 270 claims produced 727 claim-domain memberships.

TABLE 4 Claim evidence by value-outcome domain. Domains are multi-label.

Outcome domain	Total	Unsup.	Weak	Mod.	Strong	Studies
Traceability / consistency / requirements quality	115	12	62	41	0	37
Integrability / interoperability / lifecycle continuity	110	13	70	27	0	35
Reproducibility / reuse / openness	86	22	56	8	0	35
Cost / schedule / effort / efficiency / automation	85	23	35	25	2	33
Reliability / mission assurance / risk	85	20	29	36	0	33
Decision support / analysis / trade-space / feasibility	78	10	28	40	0	33
Generality / scalability / tailoring	78	20	51	7	0	34
Communication / collaboration / knowledge	58	14	38	6	0	34
Other / uncategorized value claim	22	3	9	10	0	9
Governance / ROI / project decisions	10	2	6	1	1	6

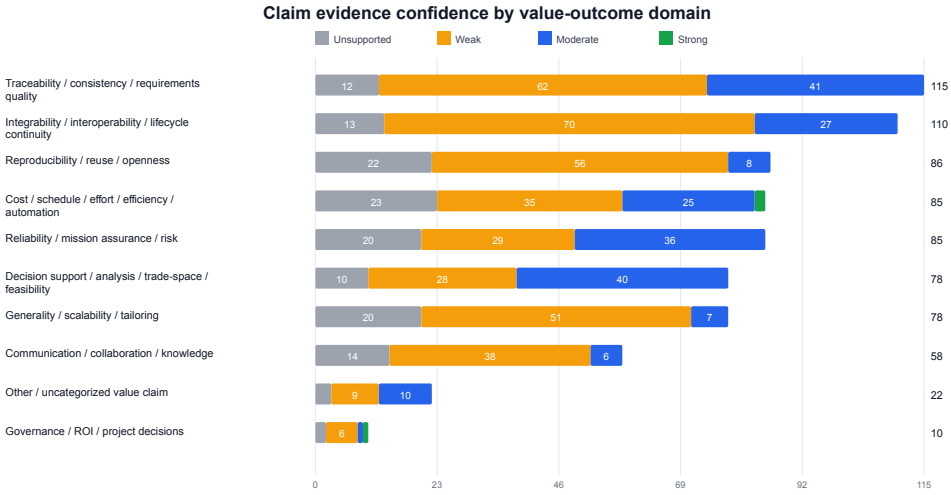


FIGURE 3 Claim evidence by value-outcome domain. Domains are multi-label, so counts exceed the number of unique claim records.

The most frequent domains were traceability/consistency/requirements quality (115 memberships across 37 studies), integrability/interoperability/lifecycle continuity (110 across 35), reproducibility/reuse/openness (86 across 35), and cost/schedule/effort/automation and reliability/mission-assurance/risk (85 each across 33 studies). EPS-specific or power-relevant claims were concentrated in integrability/interoperability (62 memberships), traceability/requirements quality (61), decision support (58), cost/schedule/effort/automation (53), reproducibility/reuse (47), and reliability/mission assurance/risk (46). The explicit "other" category reflects the open-coding policy: observations that did not fit controlled outcome labels were preserved rather than reclassified through keywords.

Reliability subtype coding made the reliability/mission-assurance domain more conservative. Model consistency or model reliability was the most frequent subtype ($n = 34$), followed by mission assurance ($n = 29$), defect/error reduction ($n = 11$), system/mission reliability ($n = 10$), process reliability ($n = 8$), generic reliability claims ($n = 2$), and EPS reliability ($n = 2$). The two EPS-reliability-coded studies were limited to cases where EPS or power behavior itself was evaluated through model-linked analysis, namely the Capella/open-source multi-domain simulation framework and the PLM/Dymola requirements-in-the-loop EPS case, rather than to broad statements that MBSE improves reliability Campioli et al. (2025); Rehman and Fortin (2024).

Moderate evidence typically came from executed simulations, model-derived calculations, traceability artifacts, or case-study outputs. Examples include the RAX model's energy/data interaction and solar-panel/battery sizing analyses Kaslow et al. (2014); Spangelo et al. (2013), Papyrus/Java power-subsystem parameter-impact and requirement-fulfillment tables Strobbe et al. (2023), power and SoC simulation against requirements in a PLM/Dymola workflow Rehman and Fortin (2024), worst-case battery survival simulation with a reported minimum state of charge in an electrical-architecture process Boan (2025), and positive mass/power margins from SysML parametrics Waseem and Sadiq (2018). Project-value metrics were rarer and case-specific, such as a one-month versus three-month feasibility-study comparison and lower estimated mission cost in one concurrent-design case Bella et al. (2024), or the reported four-month preliminary-design cycle using Arcadia and weekly sprints Buitrago-Leiva et al. (2024).

Unsupported and weak claims were common where papers asserted reduced cost, reduced schedule, improved

reliability, improved learning, or greater reuse without a baseline, metric, reuse trial, stakeholder evaluation, integration test, or flight/operational evidence. For example, the EPS-specific Papyrus study provided moderate evidence for traceability and requirement-constraint checking, but its cost/time savings remained a future quantitative-benefit study rather than an observed outcome Strobbe et al. (2023). Reliability/mission-assurance/risk claims accounted for 85 controlled outcome-domain memberships, with 36 moderate and none strong. The two strong claims were not EPS-specific reliability claims; both concerned cost/schedule effects in a single concurrent-design feasibility comparison Bella et al. (2024).

2.6 | Verification and validation linkage

For V&V, each study could receive multiple linkage categories based on controlled EPS/V&V artifact, pre-physical verification, and integration/test-readiness codes. EPS/power-specific pre-physical verification appeared in 25 studies; parametric, budget, or constraint checks in 23; executable simulation or scenario analysis before build in 22; test cases, procedures, or V&V activity packages in 21; requirements-to-model traceability or verification matrices in 19; and external tool or data-flow coupling for V&V in 11. Twelve studies had no explicit EPS-focused pre-physical V&V, 10 linked the model to physical, SIL/HIL, or integration-test readiness, and 3 had only generic or planned V&V linkage.

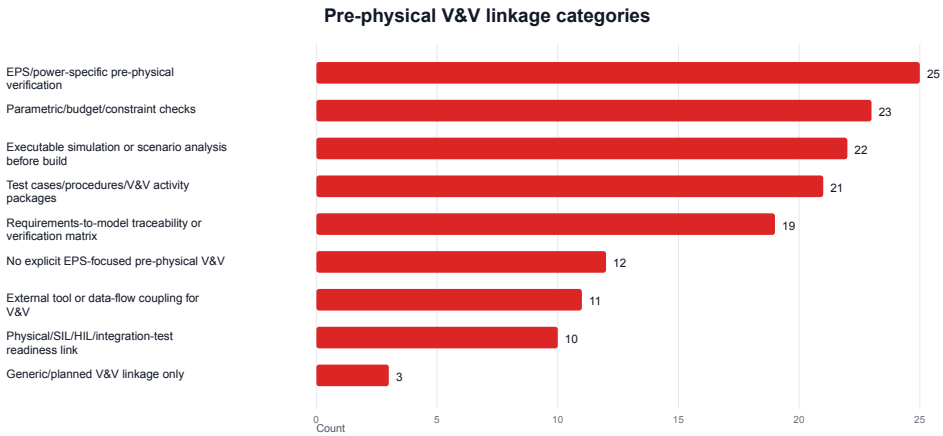


FIGURE 4 Pre-physical V&V linkage categories. Categories are multi-label by study.

TABLE 5 Pre-physical V&V linkage counts. Categories are multi-label by study.

V&V linkage category	n
EPS/power-specific pre-physical verification	25
Parametric/budget/constraint checks	23
Executable simulation or scenario analysis before build	22
Test cases/procedures/V&V activity packages	21
Requirements-to-model traceability or verification matrix	19
No explicit EPS-focused pre-physical V&V	12
External tool or data-flow coupling for V&V	11
Physical/SIL/HIL/integration-test readiness link	10
Generic/planned V&V linkage only	3

A separate IVV integration-level code recorded the highest supported integration level for each study. The most common selected level was simulation-in-the-loop (n = 12), followed by none/not reported (n = 10), executable or model-based checks (n = 6), trace-linked verification artifacts (n = 5), model-contained verification artifacts (n = 2), separate document only (n = 1), and software/hardware-in-the-loop (n = 1). Three studies had no selected integration-level value because IVV was not applicable or not reported without a controlled level assignment. This field was used as a maturity ceiling rather than as a multi-label count of every V&V mechanism.

TABLE 6 Highest coded IVV integration level.

IVV integration level	n
Simulation-in-the-loop	12
None/not reported	10
Executable/model-based checks	6
Trace-linked verification artifacts	5
Model-contained verification artifacts	2
Separate document only	1
Software/hardware-in-the-loop	1

Together, the mechanism counts and the IVV maturity coding show that V&V was usually represented through EPS/power analysis, parametric checks, simulation, and planned or modeled test artifacts. Explicit links from the model to test execution or integration evidence were still less common than pre-physical analysis, and the focused IVV coding separated actual model integration from generic V&V language.

Requirements traceability was usually represented through SysML satisfy/verify relations, verification matrices, or tool-generated traceability views. In the EPS-specific case, requirement constraints and power-budget equations were extracted from Papyrus and evaluated by Java, producing respected/not-respected flags Strobbe et al. (2023). In academic nanosatellite project modeling, requirements were linked to "Verified By" test cases and behavior diagrams such as Bus System Activation and LEOP flow Hanafi et al. (2024). In the 3ColStar preliminary design, EPS continuous-power requirements were traced to physical components such as solar panels and batteries Buitrago-Leiva et al. (2024). In the NSS-1 SysML case, verify relationships connected mass and power requirements to constraint blocks, and positive margins were used as verification evidence Waseem and Sadiq (2018).

Parametric and executable V&V evidence was strongest when requirement checks were tied to simulation or budget outputs. RAX/CubeSat studies executed mission scenarios and trade studies involving energy collection, battery capacity, solar-panel sizing, data storage, and downlink Kaslow et al. (2014); Spangelo et al. (2013); Anderson et al. (2014). Later studies linked pre-physical V&V to open-source model-to-analysis pipelines Cordero et al. (2022); Luccisano et al. (2024); Campioli et al. (2025), PLM/Dymola requirements-in-the-loop simulation Rehman and Fortin (2024), and electrical-architecture energy-balance simulations before hardware development Boan (2025). Test-case or automated-test evidence was more limited and often not EPS-specific, as in the requirements modeling/mapping matrix for CubeSat tests Alanazi et al. (2019) and the EIRSAT-1 communications verification model coupled to Python and ground-station hardware Akarapu et al. (2023). The physical/SIL/HIL/integration-test readiness category (n = 10) was broader than completed hardware testing: most linkage still stopped at readiness, planned integration, or partial software/hardware interfaces rather than closed-loop EPS acceptance evidence.

2.7 | Gap mapping

Each study was mapped to five gap categories adapted from the broader space-MBSE gap frame: practical implementation pathway; toolchain fragmentation/interoperability; lightweight/resource-constrained tailoring; governance/project deliverables/ROI/value; and NewSpace/agile/distributed/rapid iteration. Classifications were coded as addressed with evidence, addressed conceptually, mentioned only, or not addressed.

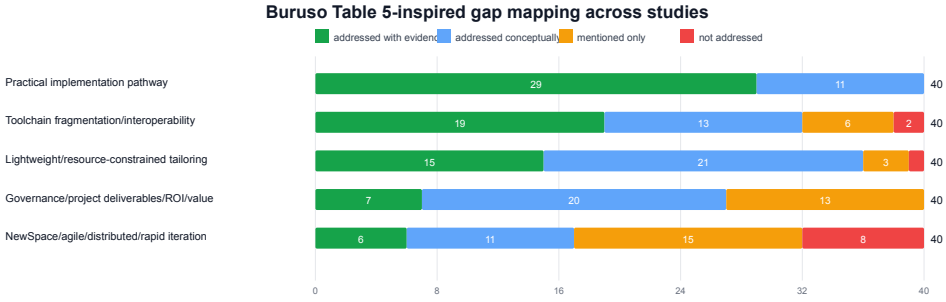


FIGURE 5 Gap-mapping counts for the extracted studies.

TABLE 7 Gap mapping counts.

Gap	Evidence	Conceptual	Mentioned	Not addressed
Practical implementation pathway	29	11	0	0
Toolchain fragmentation/interoperability	19	13	6	2
Lightweight/resource-constrained tailoring	15	21	3	1
Governance/project deliverables/ROI/value	7	20	13	0
NewSpace/agile/distributed/rapid iteration	6	11	15	8

Implementation pathways were the most consistently addressed gap: all 40 studies addressed the pathway at least conceptually, with 29 evidence-backed and 11 conceptual classifications. Evidence-backed examples included concrete modeling/data-flow steps for EPS traceability Strobbe et al. (2023), concurrent electrical-architecture simulation workflow Boan (2025), and Capella/Arcadia plus agile preliminary-design execution Buitrago-Leiva et al. (2024). Toolchain fragmentation/interoperability was also frequently addressed, with 19 evidence-backed cases, 13 conceptual cases, 6 mentioned-only cases, and 2 not-addressed cases. Evidence-backed toolchain examples included executable RAX integrations Kaslow et al. (2014); Spangelo et al. (2013), Papyrus-to-Java EPS analysis Strobbe et al. (2023), open-source model-to-simulation coupling Campioli et al. (2025), and model-to-test/tool integration in EIRSAT-1 Akarapu et al. (2023).

Lightweight/resource-constrained tailoring was addressed with evidence in 15 studies, addressed conceptually in 21, mentioned only in 3, and not addressed in 1. Evidence-backed tailoring appeared in studies with rapid analytical design-space exploration Lowe and Macdonald (2014), academic project-management and design tailoring Hanafi et al. (2024), open/concurrent preliminary-design tooling Cordero et al. (2022), docs-as-code for a university CubeSat Chiu et al. (2023), causal MBSE user evaluation Peters et al. (2022), low-cost tradespace modeling Nardi et al. (2025), and agile distributed preliminary design Buitrago-Leiva et al. (2024).

Governance, project deliverables, ROI, and value linkage remained weaker than implementation-pathway and

toolchain evidence, but the controlled recoding identified 7 evidence-backed studies, 20 conceptual studies, and 13 mentioned-only studies. Evidence-backed examples included the 3ColStar study, which reported Capella/Arcadia governance, weekly sprints, requirement/design traceability, and a four-month preliminary design cycle Buitrago-Leiva et al. (2024), as well as studies linking models to project deliverables, cost/schedule comparisons, or requirements-in-loop decision artifacts Hanafi et al. (2024); Rehman and Fortin (2024); Bella et al. (2024). NewSpace/agile/distributed/rapid-iteration evidence was also limited, with 6 evidence-backed studies, 11 conceptual, 15 mentioned-only, and 8 not addressed.

3 | DISCUSSION

3.1 | Relationship to the broader space-MBSE review

Buruso et al. provide the appropriate background for this review because their comprehensive space-MBSE synthesis already covers the broad distribution of MBSE methods, modeling languages, tools, lifecycle phases, and enabler classes across space applications Buruso et al. (2026). The present review therefore does not treat the recurrence of SysML, Cameo/MagicDraw, Capella/Arcadia, PLM tools, simulation environments, or integrated digital-engineering toolchains as a novel finding by itself. Instead, those enablers are interpreted only where they affect nanosatellite and EPS-relevant value evidence, implementation burden, or model-linked V&V.

The strongest connection to Buruso et al. is their discussion of value-driven implementation and their Table 5 research gaps. They identify persistent gaps in practical implementation pathways, fragmented toolchains and lifecycle interoperability, resource-mismatched frameworks for small missions, weak linkage between MBSE artifacts, governance and ROI, and misalignment with NewSpace-style development needs Buruso et al. (2026). Our results support those gaps in a narrower population and make their nanosatellite consequences more explicit. In the included studies, practical implementation pathways and toolchain interoperability are frequently addressed, with evidence-backed classifications in 29 and 19 studies respectively, but governance/ROI/value and NewSpace/agile/distributed development are less frequently addressed with evidence (7 and 6 studies respectively; Table 7). Thus, the nanosatellite literature is not simply missing MBSE awareness; it is missing durable evidence that the models change project outcomes, review decisions, or downstream verification practice.

This narrower focus also exposes an EPS-specific asymmetry that is hidden in broad space-MBSE taxonomies. Many studies include EPS or power concerns inside whole-CubeSat models, but only two included studies are EPS-specific or EPS-first Strobbe et al. (2023); Rehman and Fortin (2024). Most examples of power modeling are embedded in system-level trade studies, reference-model work, or preliminary design workflows such as RAX energy/data simulation Kaslow et al. (2014); Spangelo et al. (2013), SysML power-budget margins Waseem and Sadiq (2018), Capella/Arcadia preliminary EPS allocation Buitrago-Leiva et al. (2024), PLM/Dymola battery simulation Rehman and Fortin (2024), and electrical-architecture concurrent engineering Boan (2025). This extends Buruso et al.'s Table 5 concern about implementation pathways: for nanosatellites, the missing pathway is not only "how to adopt MBSE," but how to maintain an EPS/system thread from mission objectives to power requirements, architectural allocation, analysis, verification procedure, and test evidence without imposing enterprise-scale overhead.

3.2 | Value-driven MBSE remains under-evidenced for nanosatellites

The review supports Buruso et al.'s claim that MBSE value must be tied to explicit objectives, model usage, governance, and evaluation criteria Buruso et al. (2026). The nanosatellite literature reviewed here contains many value

claims, but the evidence is generally modest. RQ2 was coded as high for only 3 studies and partial for 37, and among 270 extracted value-claim records only 2 were classified as strong evidence; 77 were moderate, 139 weak, and 52 unsupported (Table 1). The most common domains were traceability, interoperability/lifecycle continuity, reproducibility/reuse, reliability/mission assurance, and cost/schedule/automation or decision-support mechanisms, but the strong evidence was confined to cost/schedule claims in one concurrent-design comparison rather than EPS reliability or verification closure (Table 4). Reliability subtyping sharpened this conclusion: EPS reliability was coded in only two studies, whereas most reliability-adjacent claims concerned model consistency, mission assurance, defect/error reduction, or process reliability.

Moderate evidence usually came from executable demonstrations or traceable artifacts rather than from measured project outcomes. For example, the RAX studies show how an integrated SysML–ModelCenter–STK–MATLAB workflow can simulate energy and data interactions, support solar-panel and battery-capacity trades, and check mission requirements Kaslow et al. (2014); Spangelo et al. (2013). The open-source Papyrus power-subsystem study demonstrates parameter-impact traceability and requirement-constraint flags from model data Strobbe et al. (2023). PLM/Dymola requirements-in-the-loop simulation reports battery state-of-charge behavior against an SoC requirement Rehman and Fortin (2024), and the electrical-architecture workflow reports a worst-case battery state-of-charge margin for survival/no-illumination conditions Boan (2025). These are useful forms of engineering evidence, but they are mostly prototype, single-case, or pre-physical demonstrations.

Evidence becomes thinner when claims move from technical feasibility to project value. Reported schedule or cost benefits exist, such as a one-month versus three-month feasibility-study comparison and lower estimated mission cost in a concurrent-design case Bella et al. (2024), or a four-month preliminary-design cycle with Arcadia and weekly sprints for 3ColStar Buitrago-Leiva et al. (2024). However, such claims are case-specific and rarely supported by controlled baselines, repeated use, independent stakeholder assessment, or lifecycle follow-through. Similarly, reference models and templates promise reuse and consistency Kaslow et al. (2020); Kaslow and Madni (2017); Kaslow et al. (2019), while docs-as-code and open-source workflows improve accessibility and reproducibility Chiu et al. (2023); Luccisano et al. (2024), but the included studies seldom demonstrate reuse across independent missions or quantify the effort required to maintain the model.

The implication is that nanosatellite MBSE remains more artifact-driven than value-driven. Studies frequently show that a model can contain requirements, architectures, budgets, interfaces, simulations, or generated documents. Fewer studies define the mission/project objective that justifies the modeling effort, the metric used to evaluate success, the baseline practice being improved, or the review gate at which the model changes a decision. This is especially important for EPS: power budgets, SoC traces, and margin checks are often convincing local analyses, but they are not always connected to explicit mission assurance objectives such as survival probability, operational availability, fault tolerance, or verification closure. In this respect, our synthesis directly supports Buruso et al.'s Table 5 gap on weak integration between MBSE artifacts, governance, ROI, and value realization, and shows that the gap persists even in a domain where resource constraints should make value justification central Buruso et al. (2026).

3.3 | V&V linkage is present but rarely continuous

The included studies do not show an absence of V&V linkage; rather, they show many partial linkages that rarely form a continuous model-to-test thread. EPS/power-specific pre-physical verification appeared in 25 studies, parametric/budget/constraint checks in 23, executable simulation or scenario analysis in 22, test cases/procedures/V&V activity packages in 21, requirements-to-model traceability or verification matrices in 19, and external tool or data-flow coupling in 11 (Table 5). The highest coded IVV integration level was usually simulation-in-the-loop, executable/model-

based checking, or trace-linked artifacts rather than model-contained verification or software/hardware-in-the-loop (Table 6). These counts indicate that nanosatellite MBSE studies often recognize verification as a modeling concern, but that the verified thread usually remains pre-physical and analytical. They also align with Buruso et al.'s observation that later lifecycle testing, integration, and verification have poorer visibility in the wider space-MBSE literature Buruso et al. (2026).

The most common V&V pattern is requirements traceability plus a local analytical check. The EPS-specific Papyrus study extracts requirement constraints and power-budget parameters to generate respected/not-respected flags Strobbe et al. (2023). The academic nanosatellite framework links requirements to "Verified By" test cases and behavior diagrams such as bus activation and LEOP flow Hanafi et al. (2024). The 3ColStar study links continuous-power requirements to EPS components and mode-level power analyses Buitrago-Leiva et al. (2024). The NSS-1 SysML case uses verify relationships and parametric constraints to compute positive mass and power margins Waseem and Sadiq (2018). These examples show useful pre-physical verification, but the verification often stops at traceability, calculation, or planned test artifacts.

Executable simulation provides a second important but still partial pattern. RAX-style workflows connect SysML behavior and parametrics to STK, MATLAB, and ModelCenter to check energy, data, and mission-scenario behavior Kaslow et al. (2014); Spangelo et al. (2013). Other studies link model data to power/data-budget simulation or mission-mode simulation Cordero et al. (2022); Luccisano et al. (2024); de Almeida et al. (2021); Campioli et al. (2025), PLM/Dymola battery behavior Rehman and Fortin (2024), and electrical-architecture energy-balance simulation Boan (2025). These studies strengthen pre-build confidence and support design trades, but they do not generally show continuity from the same model elements to formal test procedures, hardware-in-the-loop execution, integration-test data, anomaly handling, or acceptance evidence.

The few test-oriented examples reinforce this distinction. Requirements Modeling Language and RMM work links requirements, code, tests, and defects for CubeSat software-oriented examples Alanazi et al. (2019). EIRSAT-1 demonstrates a communications-subsystem verification model coupled through Python to ground-station hardware and requirements-compliance updates Akarapu et al. (2023). These are among the clearest model-to-test patterns in the included studies, but they are not EPS demonstrations. For EPS, the literature more often contains power budgets, margin calculations, and simulations than a continuous thread from EPS requirement to model allocation, analysis case, test case, test procedure, measured test data, and verification closure. This is the central V&V gap for nanosatellite MBSE: linkage exists, but it is fragmented across requirements traceability, parametric checks, simulation, and planned V&V packages.

3.4 | Implications for lightweight nanosatellite MBSE practice

For resource-constrained CubeSat and nanosatellite teams, the practical implication is not to adopt more MBSE enablers, but to adopt a thinner value-and-verification thread. Buruso et al.'s Table 5 calls for modular, scalable MBSE frameworks and adoption playbooks for lightweight missions Buruso et al. (2026). Our synthesis suggests that such playbooks should start by defining the value hypothesis for the model: which decision, review, risk, or verification activity the model is expected to improve; what artifact will be produced; what metric or comparator will indicate success; and when the modeling effort should stop. Without these criteria, teams can accumulate model artifacts while still lacking evidence that MBSE improved reliability, reproducibility, integrability, or schedule performance.

A lightweight nanosatellite playbook should make EPS a first-class model thread rather than a subsystem table appended to the architecture. At minimum, the thread should connect mission modes and operational scenarios to EPS requirements, power-generation/storage/distribution architecture, assumptions, margins, budget equations,

executable checks, verification methods, and test cases. Studies already provide pieces of this thread: open-source Papyrus/SysML traceability for a power subsystem Strobbe et al. (2023), model-derived power and data budgets Luccisano et al. (2024), Capella/PVMT simulation coupling Campioli et al. (2025), PLM/Dymola requirements-in-the-loop behavior Rehman and Fortin (2024), and concurrent electrical-architecture simulation Boan (2025). The missing practice guidance is how a small team should select a minimal subset of these mechanisms and maintain it through review and test readiness.

Tool choices should be evaluated against adoption burden as well as analytical capability. Enterprise or integrated environments can provide powerful lifecycle continuity, as shown by PLM/Dymola and concurrent-engineering workflows Rehman and Fortin (2024); Boan (2025), but they may reproduce the resource-mismatch problem noted by Buruso et al. Buruso et al. (2026). Conversely, open-source, plain-text, or docs-as-code approaches can reduce licensing and training barriers while supporting traceability and generated deliverables Strobbe et al. (2023); Luccisano et al. (2024); Chiu et al. (2023). The OPAL user study suggests that low-training conceptual-design support can be valuable for student teams Peters et al. (2022), but it also shows that usability is not equivalent to verification capability. Lightweight practice therefore needs an evidence ladder: accessibility and usability are entry criteria, while value realization requires traceable decisions, executed checks, and eventually test closure.

The strongest practice pattern in the included studies combines bounded scope, explicit roles, and recurring review cadence. The 3ColStar case links Capella/Arcadia modeling with weekly sprints, permissions, subsystem responsibilities, traceability, and preliminary-design outputs Buitrago-Leiva et al. (2024). Academic project-management frameworks similarly connect model artifacts to PBS/WBS/CBS and review gates, although they do not quantify ROI Hanafi et al. (2024). Such examples suggest that adoption playbooks should include not only modeling diagrams and tool scripts, but also governance: who owns requirements, who approves assumption changes, how power-budget changes propagate, what evidence is required at PDR/CDR-like gates, and which artifacts are frozen or regenerated for test planning.

3.5 | Threats to validity and limitations of this review

Several threats limit the strength and generality of this discussion. The search scope was limited primarily to Scopus and IEEE Xplore, with Google Scholar used only for targeted exact-title verification during recovery searches. Grey literature, internal mission documentation, tool repositories without papers, and industrial reports may contain stronger evidence for MBSE value or model-linked V&V than the accessible academic literature. Four potentially eligible reports were not retrieved or remained inaccessible and were excluded from extraction, so the synthesis reflects 40 accessible studies rather than the full set of potentially relevant work.

Extraction and synthesis also have reliability limits. The review used a predefined protocol, standardized extraction form, run-specific codebook, supporting quotations, quality appraisal, extraction-confidence ratings, and a focused per-study re-coding pass for later codebook refinements, but no independent inter-rater reliability assessment was performed. Claim-evidence confidence, EPS relevance, V&V linkage, reliability subtype, IVV integration level, and Table 5-inspired gap mappings therefore include reviewer judgment. Controlled-code mapping helped make counts reproducible and reduced the earlier keyword-categorization problem, but it can still simplify nuanced paper content, especially where authors use broad terms such as validation, verification, digital engineering, traceability, or model integration differently. Emergent codes were retained to keep such observations available for qualitative interpretation.

The evidence base is immature: all 40 studies carried at least one structured limitation flag related to incomplete lifecycle coverage, weak claim–evidence linkage, single-case or limited-scope evidence, no independent validation, and prototype/future-work maturity; 39 lacked physical-test or flight evidence; 38 lacked a baseline or comparator;

35 lacked quantitative MBSE-outcome evaluation; and 34 did not make model assets available. Repeated publications in the CubeSat System Reference Model lineage also mean that publication count should not be interpreted as independent replication. Because only two included studies are EPS-specific, EPS conclusions necessarily rely on a mixture of whole-nanosatellite, EPS-adjacent, and power-budget/electrical-architecture studies.

Use of Buruso et al.'s Table 5 gaps is another limitation. The gaps were used as a positioning and sensitizing frame, not as an independent coding validation instrument Buruso et al. (2026). Mapping nanosatellite studies onto those broad gaps is useful for comparison, but the resulting categories are coarse and should be interpreted as evidence-strength descriptors rather than definitive classifications of each study.

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